

ZUDIO RETAIL SALES PERFORMANCE

ANALYTICS DASHBOARD

Project Documentation
SQL + Power BI Retail Analytics Project

1. Introduction

This project turns raw Zudio retail data into simple, clear dashboards. It uses SQL to build and clean the database, and Power BI to create the visuals.

The data is organized into five connected tables: Stores, Customers, Products, Orders, and Order Items. The final result is four interactive dashboards, each covering a different part of the business.

2. Problem Statement

Raw retail data in tables is hard to read and understand quickly. This project solves that by turning the data into a clear, visual dashboard that anyone can use to understand business performance at a glance.

3. Objectives

- Collect and organize retail data into a clean structure
- Clean the data before building any charts
- Build a structured SQL database with connected tables
- Create DAX measures for sales, profit, orders, and products
- Build four interactive Power BI dashboards
- Find useful patterns in sales, products, stores, and customers

4. Tools and Technologies

Tool	Purpose
SQL	Build and clean the database
Power Query	Clean and prepare the data
Power BI	Build the interactive dashboards
DAX	Create measures and calculations
CSV	Store and import the dataset

5. Dataset Description

A synthetic, Zudio-style retail dataset was built for this project — realistic enough for genuine analysis, but made for learning and dashboard practice rather than pulled from a live business system.

Table	What it holds
Stores	Store location, tier, and staff details
Customers	Customer details for customer-level analysis
Products	Product category, brand, and clothing type
Orders	Order date, status, payment method, and links to customers and stores
Order Items	Item-level sales, quantity, and profit data

6. Data Cleaning and Preparation

Before any dashboard was built, the data was checked and cleaned in Power Query — incorrect or missing values would have made every chart built on top wrong too.

- Checked column data types and overall data quality
- Found and handled empty and missing values
- Found and fixed duplicate and repeated ID numbers
- Made text values consistent, including payment method names
- Checked that table relationships (primary and foreign keys) were correct
- Built a Date table for time-based analysis

7. Table Relationships

Common ID columns connect the tables, so Power BI can filter and analyze information across all of them together.

- Customers → Orders, using customer_id
- Stores → Orders, using store_id
- Orders → Order Items, using order_id
- Products → Order Items, using product_id
- Date Table → Orders, using order date

8. DAX Measures

A separate “All Measures” table keeps every DAX measure organized in one place, built using simple functions like SUM, COUNT, DISTINCTCOUNT, and DIVIDE.

Area	Examples
Sales	Total Sales, Total Profit, Average Selling Price
Orders	Total Orders, Completion Rate %, Return Rate %
Products	Total Products, Total Categories, Total Brands
Stores	Total Stores, Total Staff, Total Selling Area
Customers	Total Customers, Total Customer Cities/States

9. Dashboard 1 — Sales Overview

A high-level view of the whole business: sales, profit, and order health in one place.

Business question: How is the overall business performing?

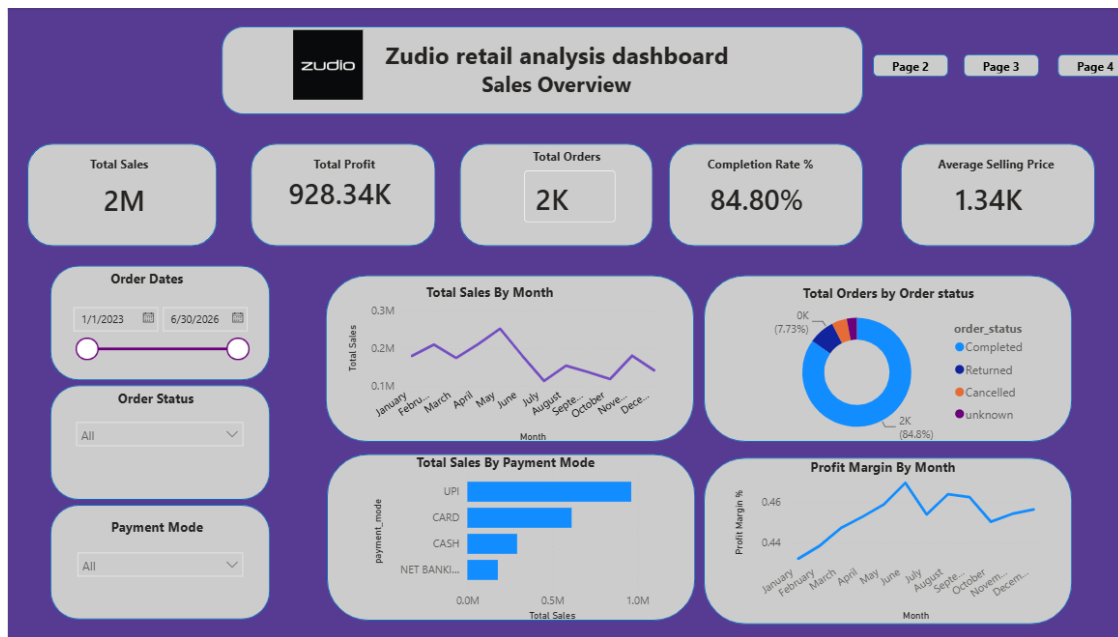


Figure 1 — Sales Overview dashboard

Measures used on this dashboard:

Total Sales • Total Profit • Total Orders • Completed Orders • Returned Orders • Cancelled Orders • Completion Rate % • Return Rate % • Average Selling Price • Total Discount • Profit Margin %

10. Dashboard 2 — Product Performance

Compares products, categories, and brands to see what is actually selling well.

Business question: Which products, categories, and brands are performing well?

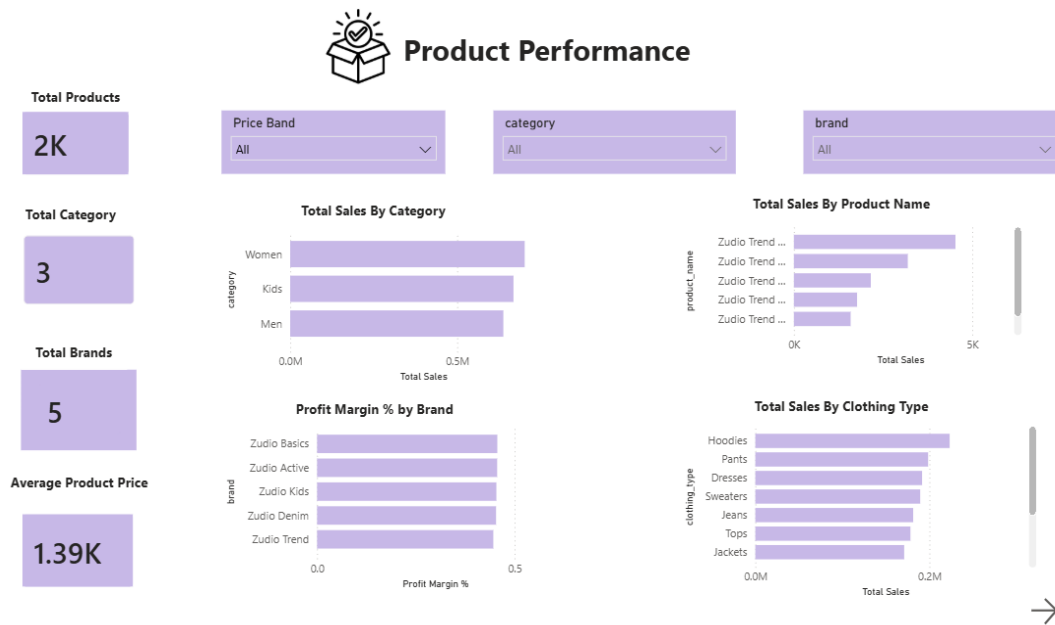


Figure 2 — Product Performance dashboard

Measures used on this dashboard:

Total Products • Total Categories • Total Brands • Average Product Price • Total Sales • Profit Margin %

11. Dashboard 3 — Store Performance

Compares stores against each other — by tier, location, staff, and sales.

Business question: Which stores are performing better, and how are store factors related to sales?

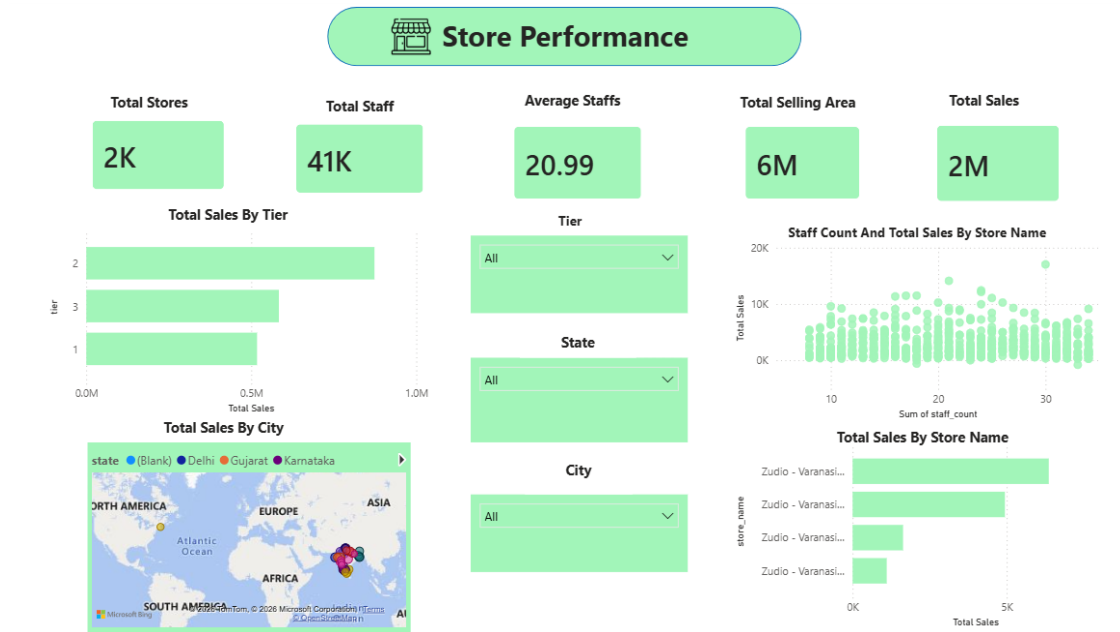


Figure 3 — Store Performance dashboard

Measures used on this dashboard:

Total Stores • Total Staff • Average Staff per Store • Total Selling Area • Total Sales

12. Dashboard 4 — Customer Insights

Shows who the customers are, where they come from, and how they shop.

Business question: Who are the customers, and how are they interacting with the business?

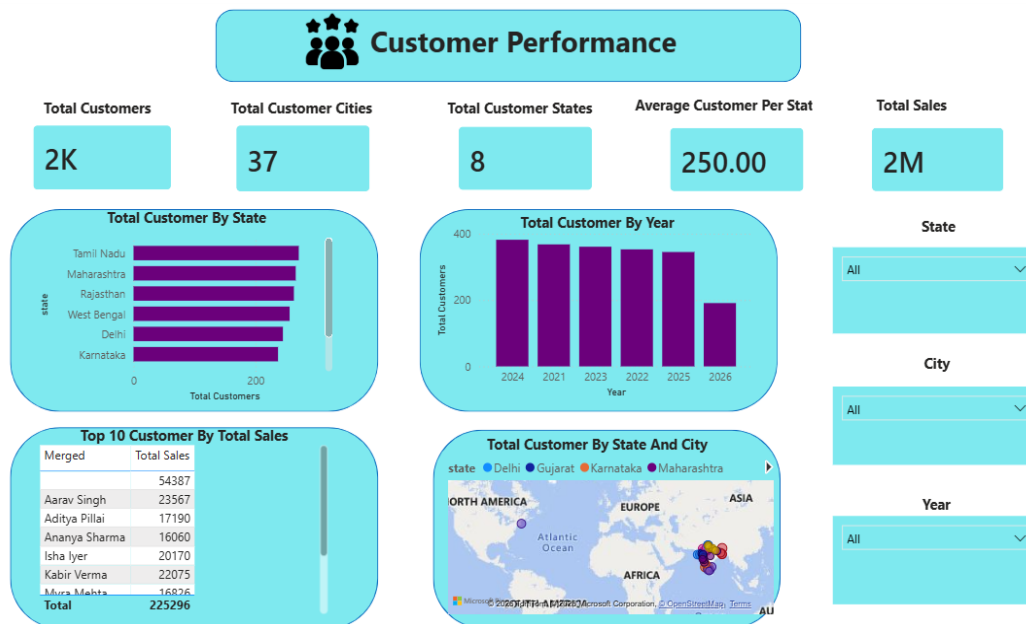


Figure 4 — Customer Insights dashboard

Measures used on this dashboard:

Total Customers • Total Customer Cities • Total Customer States • Average Customers per State • Total Sales

13. Key Insights and Business Use

The dashboards surface patterns that are hard to see in raw rows of data:

- See which products and categories bring in the most sales
- Compare high- and low-performing stores
- Track order status and payment method trends
- Watch sales and profit change over time
- Understand where customers come from and how they shop

14. Project Workflow

Collect Data → Clean Data → Build SQL Database → Connect Tables → Create Measures → Build Dashboards → Get Insights

15. Limitations and Next Steps

Limitations

- Uses sample (synthetic) data, not a live Zudio database
- Built for learning and demonstration, not live business use
- Shows current and past trends only — no forecasting yet

17. Conclusion

This project shows how SQL and Power BI work together to turn raw retail data into a clear, useful dashboard. Splitting the analysis into Sales, Products, Stores, and Customers makes it much easier to understand how the business is doing — and to make better decisions from it.

The main outcome is one interactive dashboard experience that makes retail information easy to explore, compare, and use.